Learning Objective 2.1: I can describe what a Software Defined Radio is, contrast it with a traditional hardware radio, and identify the SDR platforms used in this course (RTL-SDR, HackRF, PlutoSDR, ADALM-PHASER).

Learning Objective 2.2: I can identify the roles of the RF frontend, ADC/DAC, and host in an SDR signal chain and describe sample-rate, bandwidth, and dynamic-range trade-offs.

Use the course SDR table from the lesson page. Show your work. Write a **Documentation** line at the end (**None** if you did not collaborate).

1. **Choosing a radio.** The course SDRs are the RTL-SDR Blog V4 (0.5 to 1766 MHz, 2.56 MS/s, 8 bits, receive only), the HackRF One (1 to 6000 MHz, 20 MS/s, 8 bits, half duplex), the PlutoSDR (325 to 3800 MHz, 20 MHz instantaneous bandwidth, 12 bits, full duplex), and the ADALM-PHASER (10.0 to 10.5 GHz, digitized by a PlutoSDR).

   (a) For each task, name a course SDR that can do it, and give the specification that rules out each of the others.

   (i) Receive ship AIS at 162 MHz.
   (ii) Display the whole FM broadcast band, 88 to 108 MHz, at once.
   (iii) Transmit a 915 MHz test signal into a cable and receive it at the same time.
   (iv) Find the direction of a 10.3 GHz beacon.

   (i) The RTL-SDR or the HackRF; 162 MHz is inside both tuning ranges. The PlutoSDR starts at 325 MHz and the ADALM-PHASER works only at 10.0 to 10.5 GHz.

   (ii) The HackRF at 20 MS/s, which shows a 20 MHz window. The RTL-SDR's 2.56 MS/s shows only 2.56 MHz, the PlutoSDR cannot tune to 100 MHz, and the ADALM-PHASER works only at X band.

   (iii) The PlutoSDR, the only full-duplex radio. The HackRF transmits or receives but not both at once, the RTL-SDR cannot transmit, and the ADALM-PHASER does not cover 915 MHz.

   (iv) The ADALM-PHASER: 10.3 GHz is inside its band, its 8 elements let it find direction, and every other radio stops at 6 GHz or below.

   (b) Explain how an SDR differs from a hardware FM radio, and why the same RTL-SDR can decode both broadcast FM and aircraft ADS-B.

   A hardware radio builds its filters and demodulator as circuits for one kind of signal. An SDR only tunes, filters, and digitizes a band, and software does the filtering, demodulation, and decoding. Both 100 MHz and 1090 MHz are inside the RTL-SDR's tuning range, so receiving either one is a matter of setting the center frequency and running a different flowgraph.

2. **The mixer and I and Q.** We set a PlutoSDR's center frequency, its LO, to 433.0 MHz, and a key fob transmits at 433.92 MHz.

   (a) Find the two frequencies at the mixer output, and say which one the low-pass filter keeps.

   $$433.92 - 433.0 = 0.92 \text{ MHz}$$
   $$433.92 + 433.0 = 866.92 \text{ MHz}$$

   The low-pass filter keeps the 0.92 MHz difference and removes the sum.

   kept: 0.92 MHz

   removed: 866.92 MHz

   (b) With a single real mixer, a second transmitter would also come out at 0.92 MHz. Find its frequency, and explain how the PlutoSDR's I and Q outputs tell the two apart.

   $$433.0 - 0.92 = 432.08 \text{ MHz}$$

   The second mixer, driven by the LO shifted 90°, gives Q. In $I + jQ$ the 433.92 MHz signal rotates one way and appears at $+0.92$ MHz, and the 432.08 MHz signal rotates the other way and appears at -0.92 MHz.

   432.08 MHz

   (c) A superheterodyne receiver, like the ADALM-PHASER, mixes a 10.25 GHz signal to an IF of 2.2 GHz with an LO above the signal. Find the LO frequency, and the image: the input on the other side of the LO, at $f_{LO} + f_{IF}$, that also mixes to 2.2 GHz.

   $$f_{LO} = 10.25 + 2.2 = 12.45 \text{ GHz}$$
   $$f_{\text{image}} = 12.45 + 2.2 = 14.65 \text{ GHz}$$

   A filter ahead of the mixer must remove 14.65 GHz, or a signal there lands on top of the wanted one at the IF.

   LO: 12.45 GHz

   image: 14.65 GHz

3. **Sample rate and data rate.** A team captures a 10 MHz-wide test signal centered on 2437 MHz with a HackRF. The usable window is somewhat narrower than f_s, because the radio's filters roll off near its edges.

   (a) Find the lowest sample rate whose window just holds the signal. Then choose a rate 25% higher, to leave margin at the edges, and find the band that window covers.

   A complex window is f_s wide, so the signal needs at least 10 MS/s. With 25% margin:

   $$f_s = 1.25 \times 10 = 12.5 \text{ MS/s}$$
   $$2437 \pm 6.25 = 2430.75 \text{ to } 2443.25 \text{ MHz}$$

   lowest: 10 MS/s

   chosen: 12.5 MS/s

   band: 2430.75 to 2443.25 MHz

   (b) GNU Radio stores each complex sample in 8 bytes. At 12.5 MS/s, find the data rate through the flowgraph and the size of a 10 s File Sink capture.

   $$8 \times 12.5 \times 10^6 = 100 \text{ MB/s}$$
   $$10 \times 100 = 1 \text{ GB}$$

   rate: 100 MB/s

   file: 1 GB

   (c) The HackRF sends 8-bit I and 8-bit Q over USB. Find the USB data rate at 12.5 MS/s, and how many times larger the flowgraph's rate is.

   $$2 \text{ bytes} \times 12.5 \times 10^6 = 25 \text{ MB/s}, \qquad \frac{100}{25} = 4$$

   USB: 25 MB/s

   ratio: 4

   (d) A teammate suggests the RTL-SDR instead. Give two separate reasons it cannot make this capture.

   Its tuning range stops at 1766 MHz, below 2437 MHz. Even inside its range, its 2.56 MS/s shows only 2.56 MHz, far less than the 10 MHz signal.

4. **Bits and dynamic range.** Each ADC bit adds about 6 dB of dynamic range. A strong local station reaches the ADC at -10 dBFS, and a weak station of interest is 55 dB below it.

   (a) Estimate the dynamic range of an 8-bit and a 12-bit converter, and the level of the weak station in dBFS.

   $8 \times 6 = 48$ dB, $12 \times 6 = 72$ dB

   $-10 - 55 = -65$ dBFS

   8 / 12 bits: 48 / 72 dB

   weak: -65 dBFS

   (b) The converter's own floor sits about one dynamic range below full scale. Which converter shows the weak station, and by how many dB does it clear that converter's floor?

   The 8-bit floor is near -48 dBFS, above the -65 dBFS station, so the 8-bit radio does not show it. The 12-bit floor is near -72 dBFS, so the 12-bit radio shows it about $-65 - (-72) = 7$ dB above the floor.

   (c) On the 8-bit radio, how much can the gain rise before the strong station clips, and does that rise make the weak station visible? Explain.

   The strong station is 10 dB below full scale, so the gain can rise 10 dB. The weak station rises by the same 10 dB, to -55 dBFS, which is still below the -48 dBFS floor. Gain moves both stations together and cannot widen the 48 dB span between full scale and the floor; seeing both needs more bits or a filter that keeps the strong station out of the window.

   (d) In the L4 flowgraph, name two things on the plots that show the ADC is clipping.

   The time sink shows I and Q flattened at ± 1, and the spectrum shows new signals where nothing transmits. The strong tone also rises by less than the gain increase.

Documentation: _____